

Highly Stretchable Shape Memory Hydrogels with Silk-Fibroin Semi-Interpenetrating Networks for Drug-Delivery Microrobot

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Abstract—Shape memory hydrogels (SMHs) provide hydrophilic and stimuli-responsive platforms for biomedical devices but suffer from mechanical fragility. To address this issue, we reinforced poly (acrylic acid) (PAA) based SMHs with biocompatible silk fibroin (SF) via UV-initiated co-crosslinking. Benefited from the semi-interpenetrating networks, PAA-SF_{6%} binary SMHs exhibit remarkably improved mechanical stretchability that is more than 10-times larger comparative to neat PAA analogues without deteriorating the shape memory functionality. On this basis, a bioinspired microrobot mimicking lotus seedpod was engineered to execute programmable shape transformations—shrink to seal active substances in holes, and recover to release them. This microrobot is promising for drug delivery that can be delicately controlled by thermal and biochemical stimuli. The strategic SF incorporation not only strengthens the deformability of general SMHs, but also expands the operational scope of SMHs in advanced biomedical engineering applications.

I. INTRODUCTION

The inferior mechanical resilience of conventional shape memory hydrogels (SMHs) originates from their homogeneous polymer networks that lack effective energy dissipation mechanisms, leading to stress concentration and brittle fracture under large deformation [1]. In this work, we propose strategic binary SMHs with remarkable stretchability, which is prepared by co-crosslinking poly (acrylic acid) (i.e., PAA) and silk fibroin (i.e., SF) (Fig. 1). Despite inherent hydrophobicity mismatch between SF and PAA, the hydrogen bonding between the former's β -folded domains and the latter's polar groups (e.g., carboxyl groups) enables intermolecular physical entanglement, establishing semi-interpenetrating networks. In this hierarchical architecture, the mechanically strong β -folded domains of SF preferentially bear more load during hydrogel deformation and consolidate PAA network, effectively distributing stress and preventing catastrophic failure. Simultaneously, the covalently crosslinked PAA network maintains structural integrity and provide shape memory functionality.

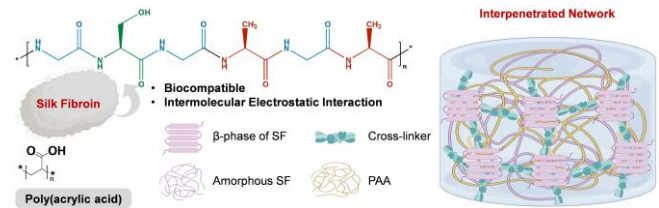


Fig. 1. Chemical structures of SF and PAA, and a schematic diagram of their composite hydrogels.

Based on PAA-SF SMHs, we further propose a biomimetic microrobot used for drug delivery [2], whose structure is inspired by lotus seedpod. As shown in Fig. 2, before shape programming, this microrobot with a diameter of 11 mm is comparable to the size of general tablets, and it incorporates multiple drug reservoirs that can be sealed through controlled compression under certain conditions. Taking advantages of shape-memory property, this design leverages inherent body temperature and body fluids to trigger drug release via sequential shape recovery.

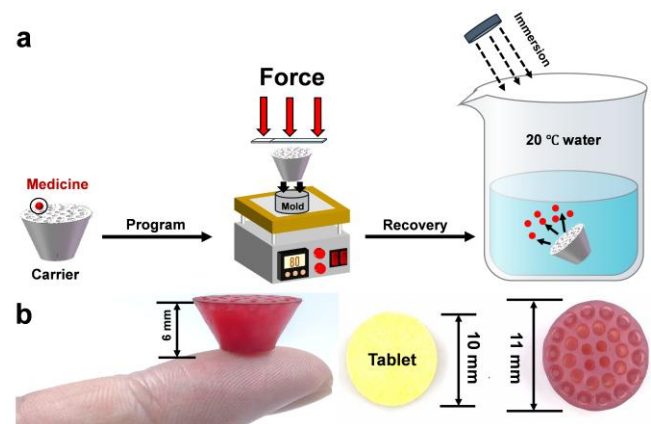


Fig. 2. Microrobot for drug-delivery. a) Flow chart for the processes of shape programming and recovery. b) 3D-printed prototype of the seedpod-structured microrobot.

II. EXPERIMENT

Fig. 3 shows the recovering process of PAA-SF hydrogel after been programmed in 20 °C IPA for 5 min. Specifically, for shape programming, a piece of PAA-SF (6 wt%)

hydrogel is bended at the angle of 90° in 20 °C isopropanol for 5 min. Then it is immersed into water at room temperature for shape recovery. During this hydration-mediated process, a critical onset period from 0 to 20 s can be observed, where the initial shape stabilization occurs due to the time required to break temporary physical crosslinks and reach critical chain mobility thresholds for network reconfiguration. Such feature of delayed recovery enables controlled drug release kinetics through gradual hydration-driven network swelling [3].

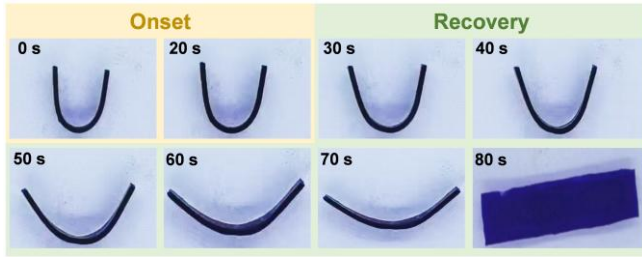


Fig. 3. Shape recovery of PAA-SF hydrogel after been programmed in 20 °C IPA for 5 min.

III. RESULTS AND DISCUSSION

Fig.4 demonstrated mechanical performance of PAA-SF SMHs with varied SF content ranging from 0 to 3, 6 and 9 wt%. Apparently, the deformability of PAA-SF SMHs drastically improved with increasing SF content. In particular, the sample with 9 wt% SF exhibits ~300% elongation at break, which is about more than 15 times larger than that of the neat PAA hydrogels. On the other hand, the fracture strength decreases with increasing SF content from 3 to 6 and 9 wt%, and the shape of PAA-SF_{9%} becomes harder to get fully recovered [4]. This might be arisen from the disruption of the chemically crosslinked PAA networks by the interposition of excessive SF pieces [5]. Overall, 6 wt% SF is found to be the optimal concentration for improving SMHs deformability, while preserving its shape memory functionality.

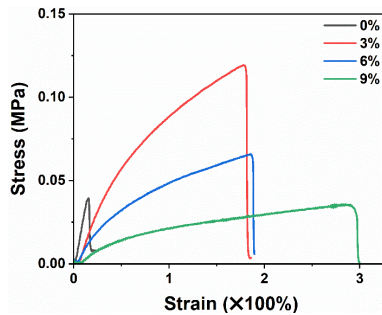


Fig. 4. Tensile stress-strain curves of PAA-SF (0, 3, 6, 9 wt%) hydrogels.

IV. CONCLUSIONS

In conclusion, we developed a highly stretchable SMHs by strategically incorporating a small fraction of SF into PAA hydrogels, leveraging molecular entanglement and SF's β -folded domains to establish semi-interpenetrated networks with remarkably improved deformability. By optimizing SF content, this hierarchical architecture enables unprecedented mechanical elongation while maintaining intact shape memory features. Finally, a seedpod-structured microrobot is designed based on this PAA-SF SMHs, which is promising for the applications of drug delivery and sustained-release. For future works, PAA-SF SMHs can hybrid with electromagnetic materials, allowing controllable device movement. The significance of this work is highlighted in demonstrating the potential of highly deformable SMHs for developing power-free microrobots.

ACKNOWLEDGMENT

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